Machine Learning 2026 • Tutorial 2

Estimating θ_1 and θ_2: Maximum Likelihood, MAP, and the Full Posterior

14 August 2026 • B2, 08:30–09:50 • Jagdish/Harsha Patel Learning Theatres

NAME (block letters)

ROLL NUMBER *one digit per box*

Today's distribution is one you have not met, and its parameters are called θ_1 and θ_2 — nothing else. That is deliberate: in this course a model's unknowns are always $\boldsymbol{\theta}$, whether the model is this curve, a logistic regression, or a network with a million weights. Nothing here can be recalled; it has to be derived. **No calculator is needed.** Work in the boxes.

Part 1 The model, and the two unknowns

You have trained a model. Now you are using it: you hand it one input, it thinks, and it gives you back an answer. That takes time — call it X, measured with a stopwatch, in milliseconds.

Do it eight times and you do not get the same number eight times. Most inputs are easy and come back quickly; the occasional awkward one takes far longer. So X is never negative, most of its values bunch up small, and a few stretch a long way to the right. The standard model for a quantity shaped like that is the **lognormal**: X is lognormal when $\log X$ is normal. Writing θ_1 for the centre and θ_2 for the variance of $\log X$,

$$p(x \mid \theta_1, \theta_2) \;=\; \frac{1}{x\sqrt{2\pi\theta_2}}\,\exp\!\left(-\frac{(\log x - \theta_1)^2}{2\theta_2}\right), \qquad x > 0.$$

Here are the eight times, in milliseconds. Because only $\log x$ appears above, both rows are given:

time to answer x_i (ms)	7.4	20.1	20.1	54.6	54.6	54.6	148.4	1096.6
$\log x_i$	2	3	3	4	4	4	5	7

log **means the natural logarithm, base** $e \approx 2.718$ **— as it does everywhere in this course, and as** `np.log` **means it in NumPy.** The eight times were chosen as $e^2, e^3, \ldots, e^7$ so that the second row is whole numbers: $e^2 = 7.39$, $e^3 = 20.09$, $e^4 = 54.60$, $e^5 = 148.41$, $e^7 = 1096.63$. If you reach for the log key on a calculator you will get base 10 and none of this will work.

(a) Look at the two rows. Why is a model for $\log X$ a more sensible thing to fit here than a model for X itself?

Part 2 Steps 1–3: one observation, all n, then logs

(a) Write $L(\theta_1, \theta_2)$ for the eight times, then $\ell = \log L$. Write $y_i = \log x_i$ to keep it short, and separate the terms that contain the unknowns from those that do not.

Part 3 Step 4 for θ_1

(a) Differentiate ℓ with respect to θ_1, set it to zero, and solve. Then evaluate.

Part 4 Step 4 for θ_2

(a) Now differentiate with respect to θ_2, set to zero, and solve. Then evaluate, using $S = \sum_i (y_i - \widehat{\theta}_1)^2$.

(b) Both parts of Step 4 used the same three moves. Name them.

Part 5 Turning the prior into a posterior

Hold θ_1 at 4. Only θ_2 is unknown now.

The problem now hands you one extra piece of information. You are not the first person to time this model. Whoever had it before you left a note:

> *"We timed it as well. The spread of* $\log X$ *came out near* 1 *— but we only ran it a handful of times."*

So two answers are on the table, and they disagree. The note says the spread is about 1. Your eight numbers say 2. Which do you believe?

Neither, on its own. The note is called a **prior**, and this part combines it with your eight numbers to get one answer, called the **posterior**.

To combine them, the prior has to be written as a curve over θ_2. Since θ_2 is a spread it can never be negative, and the usual curve for a quantity like that is the **inverse-gamma**. It carries two numbers, a and b:

$$p(\theta_2) \;\propto\; \theta_2^{-a-1} \exp\!\left(-\frac{b}{\theta_2}\right), \qquad \text{here } a = 3 \text{ and } b = 4.$$

Think of a as how much weight the belief carries, and b as the total those imagined observations bring with them. This prior carries weight $a + 1 = 4$. Today's data will carry weight $\frac{n}{2} = 4$ — for a spread, each timing counts a half. So the two sides of this contest enter *exactly tied*, on purpose: you will see the tie in the answer.[1]

This whole part in one line. Prior IG$(3, 4)$, times eight numbers, gives posterior IG$(7, 12)$ — and you get there by adding exponents, not by integrating anything.

Here is why it is that easy. The likelihood from Part 4 and the prior above are the same shape:

$$\underbrace{\theta_2^{-n/2}\, e^{-S/2\theta_2}}_{\text{likelihood}} \qquad \text{and} \qquad \underbrace{\theta_2^{-a-1}\, e^{-b/\theta_2}}_{\text{prior}}$$

Both are *(a power of* θ_2*)* $\times$ exp$(-a$ *constant*$/\theta_2)$. Multiply two things of that shape and the powers add and the constants add. Nothing else can happen.

(a) Fill the table. The prior row is done for you; use $n = 8$ and $S = 16$, and put in numbers, not symbols.

	power of θ_2	**constant inside the** exp
prior, $a = 3,\ b = 4$	$-a - 1 = -4$	$b = 4$
likelihood, $n = 8,\ S = 16$	_____	_____
product = posterior	_____	_____

Use the space below to do the multiplication itself.

[1] Most books write these two numbers α and β; they are the same two numbers. The full density carries a constant $b^a/\Gamma(a)$ in front, where Γ is the factorial extended to numbers that are not whole. You will never evaluate it here — everything below is done up to proportionality, so it cancels.

(b) An inverse-gamma(a, b) has power $-a-1$ and constant b. Read your product row against that and name the posterior.

(c) Your posterior is IG$(7, 12)$. The mode of an inverse-gamma(a, b) is $b/(a+1)$ and its mean is $b/(a-1)$. Give the MAP estimate and the posterior mean, and say why they are not the same number.

(d) You are holding a whole distribution, not a number: its central 95% runs from about 0.9 to 4.3. So which one number should the model use — MLE 2.0, MAP 1.5, or posterior mean 2.0?

Where we end today — two things to take with you, neither of which you have to derive.

1. This pattern has a name, and you have seen it before. A prior that comes back in its own family after meeting the data is called **conjugate**, and in every case the prior's parameters are counts and totals to which the data simply adds its own:

Data model	Unknown	Prior	Posterior
Bernoulli	a proportion	Beta(a, b)	Beta$(a + k,\ b + n - k)$
Normal, spread known	the centre	Normal(m, s^2)	Normal, both updated
Normal, centre known	**the spread**	**Inv-gamma**(a, b)	**Inv-gamma**$\big(a + \frac{n}{2},\ b + \frac{S}{2}\big)$
Poisson	a rate	Gamma(a, b)	Gamma$(a + \sum x_i,\ b + n)$

Lecture 3's Beta prior was the first row. Today was the third.

2. MAP is regularisation. Take logs of the MAP objective and flip the sign:

$$\underbrace{\arg\max_{\boldsymbol{\theta}} \big[\log p(x \mid \boldsymbol{\theta}) + \log p(\boldsymbol{\theta})\big]}_{\text{MAP}} = \arg\min_{\boldsymbol{\theta}} \big[\underbrace{-\log p(x \mid \boldsymbol{\theta})}_{\text{loss}} + \underbrace{-\log p(\boldsymbol{\theta})}_{\text{penalty}}\big].$$

Fitting with no penalty is the MLE — least squares is exactly this. Fitting with one is the MAP, and when the model is a regression the penalty has the names you will meet later: a Gaussian prior on the weights gives ridge, a Laplace prior gives lasso, and the prior's strength is λ. The prior here carried weight 4, exactly what eight timings carry, which is why the answer landed exactly halfway between what the prior said and what the data said — and why, on eight hundred timings, the prior would hardly move it at all.

Later in this course: *This is the difference between fitting a model and knowing how much to trust it.*

The four steps used today are the four from Lecture 3: write $p(x_i \mid \boldsymbol{\theta})$, multiply over the n observations, take logs, then maximise — by calculus where that works, and by argument where it does not. **After the tutorial:** worked solutions are released on the portal, with the two profile curves and the prior–likelihood–posterior picture drawn out. The recipe itself is always available at `ml2026.floodresq.com/explore/mle`.